

AUGHTON, United Kingdom

WMO: 033220

Lat: 53.550N

Lon: 2.917W

Elev: 56

StdP: 100.65

Time Zone: 0.00 (EUW)

Period: 86-96

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-2.9	-1.5	-6.4	2.2	-1.4	-4.8	2.5	-0.3	14.5	7.1	12.7	6.9	3.4	120	0.633

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.0	24.5	17.4	22.4	16.7	20.5	15.9	18.3	23.0	17.4	21.2	16.6	19.7	4.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.7	12.0	19.4	15.8	11.3	18.7	15.1	10.8	18.1	51.8	23.1	49.1	21.1	46.6	19.7	21.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.5	10.2	9.1	DB	-4.7	27.5	2.4	2.6	-6.5	29.4	-7.9	30.9	-9.3	32.4	-11.1	34.3	(4)
(5)			WB	-5.3	20.0	2.3	0.8	-7.0	20.6	-8.4	21.0	-9.7	21.5	-11.3	22.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.6	4.6	4.4	6.3	8.0	11.4	13.6	15.9	15.3	12.9	10.4	7.2	5.3	(6)
(7)		DBStd	4.86	3.08	3.18	2.51	2.75	2.88	2.77	2.34	2.56	2.14	2.66	2.85	3.18	(7)
(8)		HDD10.0	802	167	157	117	74	16	2	0	0	2	28	91	147	(8)
(9)		HDD18.3	3193	425	390	373	311	215	146	83	101	163	247	335	403	(9)
(10)		CDD10.0	668	1	1	2	13	59	109	182	165	89	39	6	2	(10)
(11)		CDD18.3	18	0	0	0	0	0	3	7	8	0	0	0	0	(11)
(12)		CDH23.3	121	0	0	0	0	3	23	36	57	2	0	0	0	(12)
(13)		CDH26.7	19	0	0	0	0	0	2	3	14	0	0	0	0	(13)
(14)	Wind	WSAvg	4.5	5.5	5.0	5.4	4.6	4.2	3.9	4.0	3.8	4.0	4.1	4.4	4.8	(14)
(15)	Precipitation	PrecAvg	1070	103	78	67	61	68	74	81	97	95	115	112	120	(15)
(16)		PrecMax	1416	200	177	152	105	118	173	178	202	195	261	227	270	(16)
(17)		PrecMin	825	14	12	22	20	14	27	27	25	15	39	55	32	(17)
(18)		PrecStd	162	44	39	33	24	29	37	34	36	39	49	45	55	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.4	12.8	14.8	20.0	23.5	26.4	26.9	28.7	22.7	18.3	14.2	13.8	(19)
(20)			MCWB	10.3	10.7	11.2	14.4	16.4	17.6	18.6	18.6	17.4	15.0	12.4	11.7	(20)
(21)		2%	DB	11.1	11.3	12.1	16.5	21.0	23.1	24.0	24.2	19.4	16.5	13.2	12.3	(21)
(22)			MCWB	9.3	9.2	9.7	12.2	15.0	17.1	17.5	17.3	15.4	13.9	11.7	10.7	(22)
(23)		5%	DB	10.0	10.3	11.1	14.0	18.0	20.3	22.1	21.3	17.6	15.3	12.1	10.9	(23)
(24)			MCWB	8.5	8.4	9.2	10.6	13.0	15.6	16.9	16.5	14.3	13.1	10.9	9.7	(24)
(25)		10%	DB	9.0	8.9	10.1	12.2	15.9	18.3	20.1	19.1	16.3	14.1	11.2	9.9	(25)
(26)			MCWB	7.6	7.3	8.4	9.4	12.1	14.7	16.2	15.5	13.5	12.3	10.1	8.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.8	10.9	11.5	14.4	17.2	19.0	19.4	19.2	17.7	15.8	13.0	12.0	(27)
(28)			MADB	12.0	12.4	13.7	18.9	22.1	24.2	24.2	25.7	22.3	17.7	14.0	13.3	(28)
(29)		2%	WB	9.6	9.6	10.2	12.6	15.3	17.5	18.4	18.1	16.1	14.4	11.9	11.1	(29)
(30)			MADB	10.7	11.1	11.7	16.0	19.7	21.8	22.7	22.8	18.8	16.0	12.9	12.3	(30)
(31)		5%	WB	8.7	8.6	9.4	11.1	13.7	16.2	17.4	17.2	15.0	13.4	11.1	9.9	(31)
(32)			MADB	9.8	10.0	10.8	13.3	17.2	19.5	20.9	20.6	16.9	14.7	12.0	10.8	(32)
(33)		10%	WB	7.8	7.6	8.6	9.9	12.5	15.0	16.5	16.2	14.1	12.7	10.2	8.9	(33)
(34)			MADB	8.9	8.8	9.9	11.9	15.4	17.9	19.5	18.6	15.8	13.9	11.1	9.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	3.9	4.3	4.8	5.9	6.6	6.3	6.0	5.9	5.7	4.9	4.2	3.7	(35)
(36)			MCDBR	5.1	5.3	5.6	8.6	10.3	10.0	9.6	9.9	7.6	6.2	4.4	4.6	(36)
(37)			MCWBR	4.8	4.4	4.2	5.1	5.5	5.6	5.0	4.7	4.5	4.4	3.8	4.5	(37)
(38)		5% WB	MCDBR	4.8	5.0	5.3	7.9	9.7	8.8	8.1	8.2	7.2	5.4	4.5	4.4	(38)
(39)			MCWBR	4.8	4.4	4.3	5.1	5.6	5.5	4.7	4.5	4.8	4.3	4.2	4.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.326	0.339	0.368	0.390	0.394	0.391	0.392	0.378	0.365	0.348	0.325	0.314	(40)	
(41)		taud	2.320	2.382	2.316	2.275	2.309	2.343	2.346	2.400	2.437	2.443	2.386	2.293	(41)	
(42)		Ebn at Noon	631	757	808	838	854	859	851	843	808	738	634	564	(42)	
(43)		Edh at Noon	60	77	100	118	120	118	115	103	88	71	56	51	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.62	1.31	2.47	3.82	4.88	5.09	4.83	3.96	2.81	1.56	0.76	0.47	(44)	
(45)		RadStd	0.06	0.14	0.19	0.34	0.36	0.46	0.47	0.25	0.21	0.13	0.08	0.05	(45)	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
(47) Regional (68 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A			

Nomenclature: See separate page